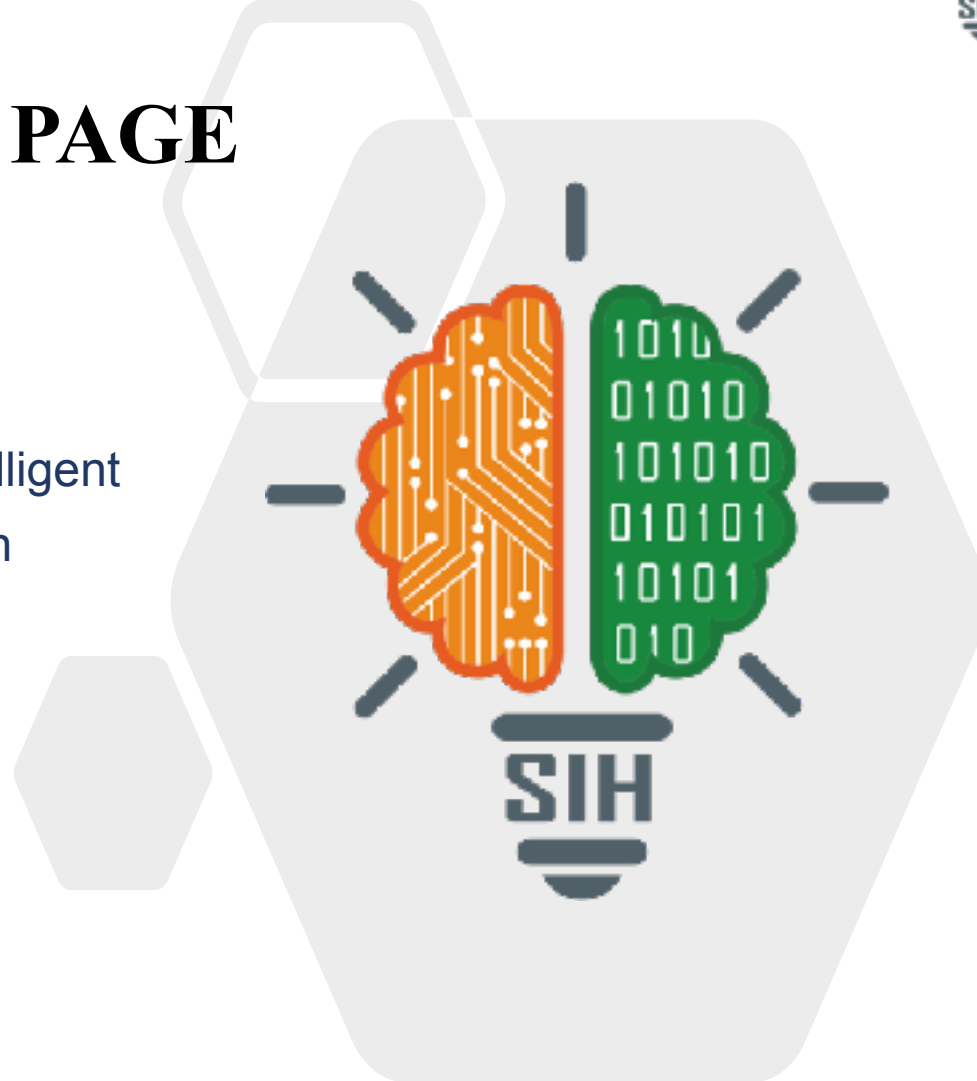


# SMART INDIA HACKATHON 2026



## TITLE PAGE

- **Problem Statement ID** – SIH26168
- **Problem Statement Title** – AI-ML based Intelligent Dead Reckoning system for seamless navigation
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – <Team ID from portal>
- **Team Name (Registered on portal)** – KANABI



# IDEA: Intelligent Dead Reckoning (IDR) + GNSS Fusion

## Proposed solution

- An edge-deployable engine + Android app that keeps lane-level position when GNSS is lost (tunnels, underpasses, parking, urban canyons, jamming) using only the phone's accelerometer, gyroscope, magnetometer and an offline OpenStreetMap graph. No OBD-II, no wheel sensor, no fixed mount.
- Hybrid AI + physics pipeline: in-vehicle alignment & calibration → AI speed/vibration filter (1-D CNN) → bias-corrected INS → map-matching with non-holonomic constraints → AI-augmented error-state Kalman fusion with instant GNSS-deficit switching.
- Training happens on desktop (IO-VNBD); the lightweight model (32 k params, ONNX/TFLite) runs on the phone at 10 Hz and on the edge engine at ~200 Hz for external/FOG IMUs.

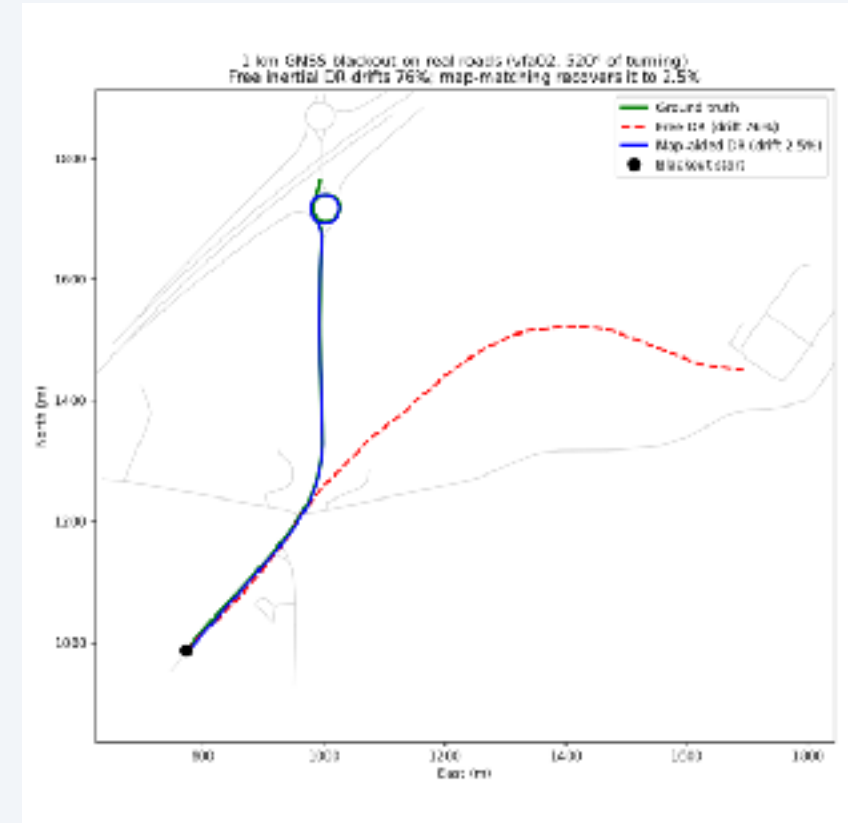
## How it addresses the problem

- GNSS drop → seamless switch to inertial tracking within ms; GNSS return → re-converge to GNSS+INS.
- Speed without OBD: last-GNSS speed anchor + learned IMU corrections; vibration, potholes and idling filtered out.
- Heading drift (the real killer of phone DR) is cancelled by snapping to the road graph on every edge.

## Innovation and uniqueness

- Data-driven design: on IO-VNBD we measured that speed holds to 1–2 % but gyro heading drifts 19–28° in 30–60 s, so we spend the AI budget where the error really is.
- Mount-agnostic calibration (gravity + PCA of driving acceleration) and one shared core engine for phone and edge/FOG sensors.

### Preliminary result on IO-VNBD (drive vfa02): 1 km GNSS blackout, 520° of turning



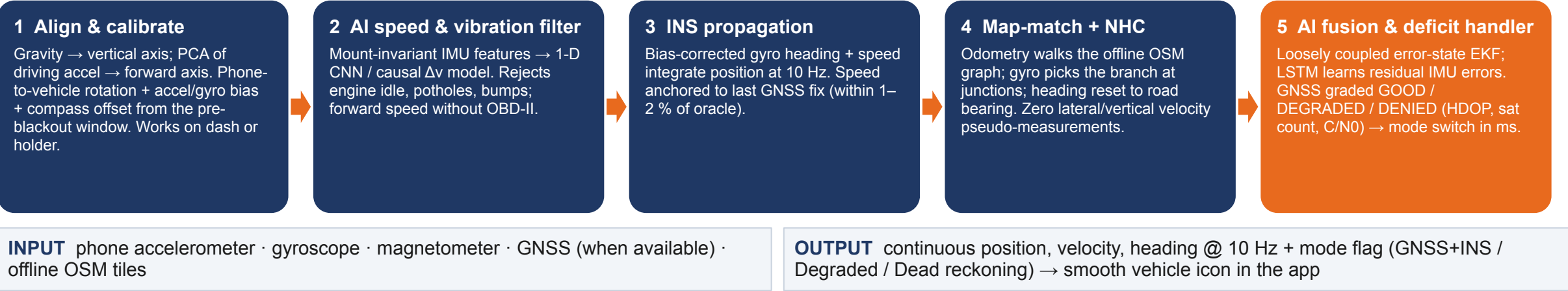
**Plain phone IMU drifts 76 %** of distance; **our map-aided IDR finishes 2.5 % off** and tracks the roundabout. Benchmark: < 10 %.



Technologies



Methodology: on-device pipeline (10 Hz on phone, ~200 Hz on edge engine)



Workflow and current status

**Phase 1 (desktop, done):** IO-VNBD loader & synchronisation audit, feature engineering, speed-filter & LSTM training, simulated-blackout evaluation over 369 one-km blackouts, OSM map-matching, browser replay demo.

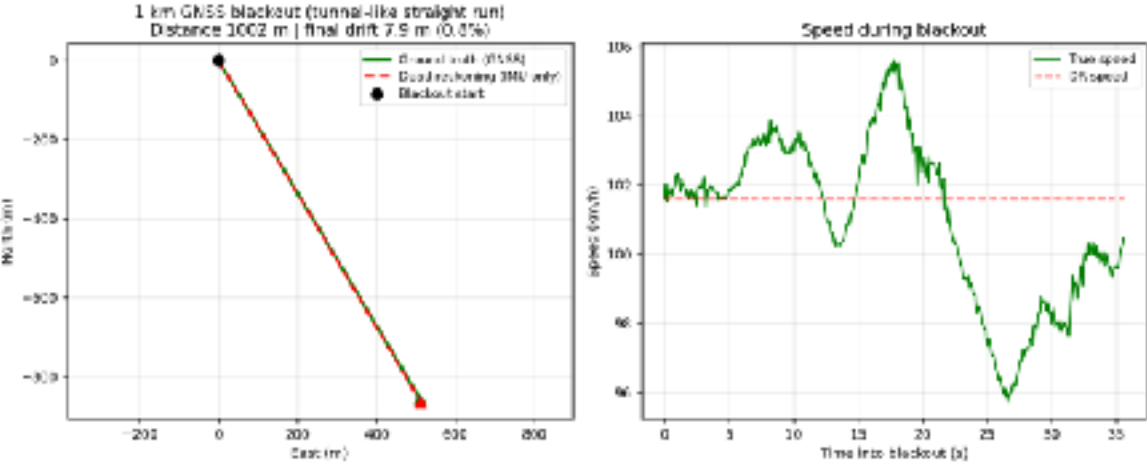
**Phase 2 (on-device, in progress):** Flutter app with live GNSS/IMU HUD and mode banner is built (APK); engine export to TFLite and porting of alignment + DR to the phone are next, followed by the 200 Hz edge build.





Feasibility: measured on IO-VNBD, smartphone IMU only

| Scenario (simulated GNSS blackout on real drives)        | PS benchmark | Achieved                    |
|----------------------------------------------------------|--------------|-----------------------------|
| 50 m blackout, motorway (< 1 min)                        | < 5 m        | 0.4 m (0.7 %)               |
| 1 km straight, tunnel-like run @ ~100 km/h               | < 100 m      | 7.9 m (0.8 %)               |
| 1 km real roads with turns, map-aided (median of 369)    | < 10 %       | 3.3 % hwy · 4.8–7.0 % urban |
| Share of 1 km blackouts under 10 % (free DR → map-aided) | —            | 17 % → 74 %                 |



Position plot inferred from an IO-VNBD test drive: 1 km blackout, final drift 7.9 m. Trained within-vehicle (Driver E), tested on held-out drives vfa02 / vt5 / vw2. Reproducible: python -m src.idr.evaluate / evaluate\_mapmatch.

Compute: 32 k-parameter CNN + EKF runs comfortably at 10 Hz on a mid-range Android phone; OSM graph for a city fits in a few MB offline.

Challenges, risks and how we handle them

| Challenge / risk                                                     | Strategy                                                                                              |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Gyro heading drifts 19–28° in 30–60 s → free DR > 10 % beyond ~200 m | Map-matching + NHC reset heading on every road edge; ZUPT at stops; magnetometer only as a weak prior |
| Absolute speed from 10 Hz vibration aliases (> 5 Hz Nyquist)         | Anchor to last GNSS speed + learned Δv correction; 100–200 Hz sampling on phone/edge for the filter   |
| Wrong branch at ambiguous junctions (remaining urban misses)         | Multi-hypothesis HMM map-matching with gyro-turn likelihood; prune on GNSS return                     |
| Phone mount changes / hand-held, two-wheelers                        | Continuous re-alignment on detected mount change; vehicle-class model heads                           |
| Different IMUs (phone vs FOG) and finale datasets                    | Sensor-agnostic core; calibration stage absorbs bias/scale; same code path for edge engine            |
| On-device latency and battery                                        | Tiny CNN, TFLite int8, 10 Hz loop; heavy training stays on desktop                                    |

Viability

- Zero hardware cost: runs on phones already on every dashboard; no OBD dongle, no vehicle integration.
- Open data (IO-VNBD) + open maps (OSM) → no licensing barrier; models retrain as finale datasets arrive.
- Working repo, reproducible pipeline, replay demo and Android APK already exist.

## Who it helps

### Delivery & ride-hailing riders

Two-wheelers and cabs with only a phone: no frozen map in flyovers, underpasses and metro tunnels.

### Trucks, fleets & older cars

Vehicles without factory INS or OBD feed get continuous tracking and correct turn-by-turn guidance.

### Emergency responders

Ambulances and fire units keep position in parking decks and canyons where seconds matter.

### Navigation & telematics apps

Drop-in engine for MapmyIndia / fleet SDKs; edge engine for ISRO, defence and rail use with FOG IMUs.

## Benchmark the judges set vs. what we already measure

**< 10 %**

PS target drift

**0.7 %**

50 m blackout

**0.8 %**

1 km tunnel run

**3.3–7 %**

1 km real roads  
(median)

## Benefits

### Social / safety

No missed exits or erratic jumps in tunnels; safer lane guidance for two-wheelers; reliable emergency response in GNSS-denied places.

### Economic

Zero added hardware; fewer failed deliveries and detours; one engine for millions of existing phones and fleet devices.

### Environmental

Fewer wrong turns and re-routing loops → less fuel and emissions per trip.

### Strategic

Resilience against jamming/spoofing and signal loss; complements NavIC; same AI core deploys on edge hardware for space, defence and rail.



### Live replay demo + Android app

Real IO-VNBD drive through a 1 km blackout: plain IMU 942 m off, IDR + map-matching 7 m. App shows a smooth vehicle icon with an unmistakable GNSS / Dead-reckoning banner.

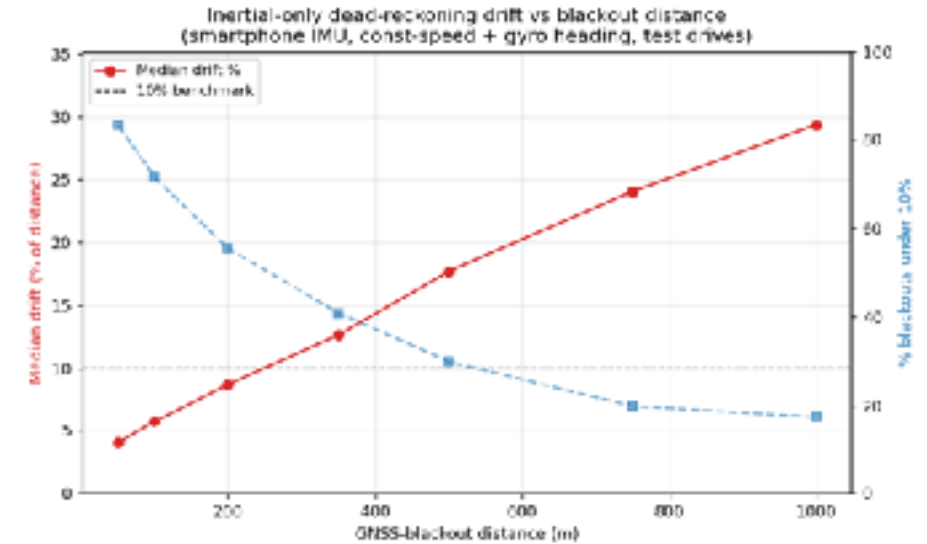
## Our work (open for review)

- **Code, models, results:** <https://github.com/AnushtupGhosh5/sih>
- Reproducible pipeline (requirements.txt / Docker): IO-VNBD loader → speed filter training → blackout evaluation → map-matching → figures; RESULTS.md holds every number on this deck.
- Browser replay demo (demo/index.html) and Android APK (mobile\_app/ui) included in the repository.

## References

- Onyekpe U., Palade V., Kanarachos S., et al. "IO-VNBD: Inertial and Odometry benchmark dataset for ground vehicle positioning." Data in Brief, 2021. [github.com/onyekpeu/IO-VNBD](https://github.com/onyekpeu/IO-VNBD)
- Newson P., Krumm J. "Hidden Markov Map Matching Through Noise and Sparseness." ACM SIGSPATIAL GIS, 2009.
- Brossard M., Barrau A., Bonnabel S. "AI-IMU Dead-Reckoning." IEEE Trans. Intelligent Vehicles, 2020.
- Herath S., Yan H., Furukawa Y. "RoNIN: Robust Neural Inertial Navigation in the Wild." ICRA, 2020.
- Solà J. "Quaternion kinematics for the error-state Kalman filter." arXiv:1711.02508, 2017.
- Groves P. D. Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems, 2nd ed., Artech House, 2013 (NHC, ZUPT, loosely coupled INS/GNSS).
- OpenStreetMap contributors, [openstreetmap.org](https://openstreetmap.org); Boeing G. "OSMnx" (Computers, Environment and Urban Systems, 2017).
- TensorFlow Lite / ONNX Runtime documentation for on-device inference.

## Key findings from our IO-VNBD study



- Inertial-only phone DR stays under 10 % only to ~150–200 m; map constraints extend this to 1 km (median 29.3 % → 3.3–7.0 %).
- Holding the last GNSS speed is within 1–2 % of oracle speed over a tunnel; gyro heading, not speed, dominates error.
- Inertial alignment of the dataset (phone vs ECU, 20 sessions) raised  $\Delta v$  correlation from 0.08 to 0.31, proving recoverable motion information in the phone IMU.
- Dataset audit found and fixed a gyro/gravity axis-matching bug that corrupts heading in naive loaders.